

### The Utility Problem

With the ever-increasing presence of climate change, extreme weather events featuring extreme temperatures are occurring more frequently. In addition, with the increasing presence of wildfires over recent years, grid assets can experience much higher temperatures. Grid assets will often have operating temperature ranges in which they will operate most efficiently and safely. If the temperature reaches certain thresholds (different for different grid assets), it could result in the following: operate inefficiently, cause an outage, cause a wildfire and/or cause damage to other grid assets.

Mechanical and electrical components are designed to operate within certain conditions, and when these conditions are not met, the components will operate at much lower efficiencies. This not only causes reductions in profits for the utility but also results in more emissions as more energy needs to be produced to meet the same demands of consumption. If the temperature is extreme enough, this can cause the grid asset to fail. There are multiple potential impacts if a grid asset breaks, such as causing an outage which will cause losses for the utility and reduce customer satisfaction. Additionally, failure could result in damage to another grid asset which would again cause losses or cause a wildfire by a spark, etc.

Overall, the impacts of extreme temperature on grid components can range from financial losses, increased emissions and customer dissatisfaction to wildfires, extended outages and other damaged assets. With this in mind, grid temperature monitoring should be a priority for utilities as extreme weather events and wildfires become more common. It is important to note that many utilities are installing more and more temperature-measuring IoT sensors, which has given some awareness in the grid, but a more in-depth analysis could prove very beneficial.

### The Data Solutions and Methods

To describe the methodology of this analysis in general terms, it is necessary to mention the required data sets which are used for this type of analysis. These datasets are:

1. Weather data corresponding to the desired locations on the distribution grid.
2. IoT sensor temperature readings.
3. Grid Asset specification data, including the operating temperature range.

This analysis can be completed efficiently in both our front-end application, True Grid Intelligence (TGI), and with Awesense Energy Data Model (EDM), accessible through our SQL EDM API. In EDM, the comprehensive data model and data access functions are the main aspects making such analysis easy to deliver. The analyst will first query all of the weather data along with any available IoT sensor temperature readings for their given service territory through our EDM API. Additionally, the analyst will then query for the operating temperature range for each grid component. The analysis can then be automated to query every hour for the temperature and weather data and compare the GIS location temperature with the location of each grid component and their corresponding rating. An automated alert can be built in order to

## Grid Temperature Monitoring

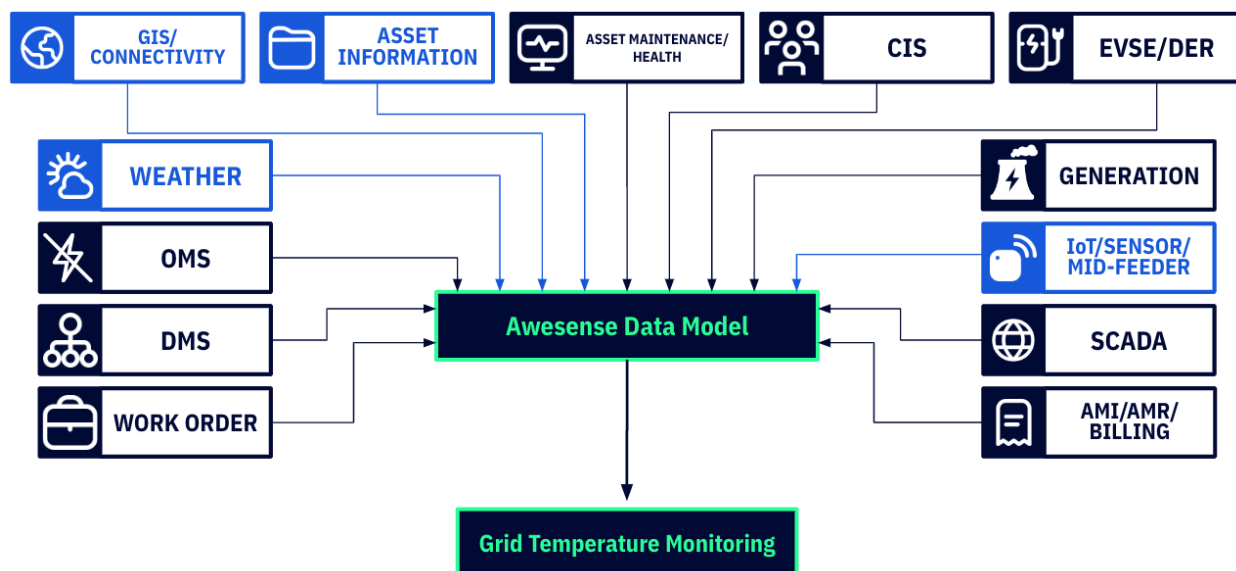
### [UC12-01] - Awesense Energy Transition Platform Use Case



flag the utility operations team when temperature ratings are exceeded. These results will provide the operation and planning teams with the information needed to change their operations to accommodate for the extreme temperatures and/or upgrade the grid components accordingly.

#### Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



#### Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.
2. TGI Grid Alerts - Please speak to Awesense about access to the sandbox environment or if you would like to see a demo.